We first demonstrate the idea of [220] by considering the following SDE with a single degree of freedom:

$$\mathrm{d}q = p\mathrm{d}t, \tag{8.40}$$

$$\mathrm{d}p = [F(q) - \lambda p]\,\mathrm{d}t, \tag{8.41}$$

$$\mathrm{d}\xi_1 = -\lambda\xi_1\mathrm{d}t - \xi_2\xi_1\mathrm{d}t, \tag{8.42}$$

$$\mathrm{d}\xi_2 = \mu_2^{-1}(\mu_1\xi_1^2 - \mathrm{k_B T})\mathrm{d}t - \gamma\xi_2 dt + \sigma\mathrm{d}W, \tag{8.43}$$

where ξ_1 and ξ_2 are auxiliary variables, $\sigma = \sqrt{2\gamma/\beta\mu_2}$, γ is the friction constant, μ_1 and μ_2 are coupling coefficients, and λ is a Lagrange multiplier which is chosen to maintain the isokinetic relation involving p and ξ_1:

$$p^2 + \frac{1}{2}\mu_1\xi_1^2 = \mathrm{k_B T}. \tag{8.44}$$

We calculate the multiplier by differentiating the constraint to obtain

$$2p\dot{p} + \mu_1\xi_1\dot{\xi}_1 = 0$$

or, using the differential equations,

$$2(pF(q) - \lambda p^2) + \mu_1\xi_1(-\lambda\xi_1\mathrm{d}t - \xi_2\xi_1) = 0,$$

thus

$$\lambda = \frac{pF(q) - \mu_1\xi_1^2\xi_2/2}{p^2 + \mu_1\xi_1^2/2}.$$

Upon reintroduction of this expression in (8.40)–(8.43) this allows us to write a closed-form SDE system for q, p, ξ_1, ξ_2.

We follow an analogous procedure to that used to study the distribution in the case of the standard isokinetic system, for the moment *ignoring the stochastic terms*. This requires us to solve the equation

$$\nabla w \cdot \boldsymbol{f} = \mathrm{div}(\boldsymbol{f}),$$

where now $\boldsymbol{f}$ is the vector field in (8.40)–(8.43) where we take $\gamma = 0$. The divergence is

$$\mathrm{div}(\boldsymbol{f}) = -\frac{\mathrm{d}}{\mathrm{d}p}(\lambda p) - \frac{\mathrm{d}}{\mathrm{d}\xi_1}(\lambda\xi_1) - \xi_2.$$

Let $\eta_1 = pF - \mu_1\xi_1^2\xi_2/2$, $\eta_2 = p^2 + \mu_1\xi_1^2/2$, so $\lambda = \eta_1/\eta_2$, then calculate

$$\frac{\mathrm{d}}{\mathrm{d}p}\lambda = (\eta_2 F - 2\eta_1 p)/\eta_2^2,$$

$$\frac{\mathrm{d}}{\mathrm{d}\xi_1}\lambda = (-\mu_1\xi_1\xi_2\eta_2 - \mu_1\xi_1\eta_1)/\eta_2^2,$$

to see that

$$\begin{aligned} p\frac{\mathrm{d}\lambda}{\mathrm{d}p} + \xi_1\frac{\mathrm{d}\lambda}{\mathrm{d}\xi_1} &= \frac{(pF - \mu_1\xi_1^2\xi_2)\eta_2 - (2p^2 + \mu_1\xi_1^2)\eta_1}{\eta_2^2} \\ &= 2\left[-\frac{pF}{2\eta_2} + \lambda - \lambda\right] = -\frac{pF}{\eta_2}, \end{aligned}$$

hence

$$\begin{aligned} \mathrm{div}(\boldsymbol{f}) &= \frac{pF}{\eta_2} - 2\lambda - \xi_2 \\ &= \frac{pF - 2(pF - \mu_1\xi_1^2\xi_2/2)}{p^2 + \mu_1\xi_1^2/2} - \xi_2 \\ &= \frac{-pF + \mu_1\xi_1^2\xi_2}{p^2 + \mu_1\xi_1^2/2} - \xi_2. \end{aligned}$$

We find that a solution of the equation $\nabla w \cdot \boldsymbol{f} = \mathrm{div}(\boldsymbol{f})$ is then given by

$$w = \frac{U + \frac{1}{2}\xi_2^2}{p^2 + \frac{1}{2}\mu_1\xi_1^2}.$$

Integrating over the isokinetic surface $p^2 + \frac{1}{2}\mu_1\xi_1^2 = \beta^{-1}$, and with respect to the variable ξ_2, yields the Gibbs configurational density.

The exciting prospect here is that we now have a new variable ξ_2 which has, in equilibrium, a Gaussian distribution. The Ornstein-Uhlenbeck terms are therefore compatible with this extended distribution and can be incorporated (as in Nosé-Hoover-Langevin) to provide ergodicity and control of the convergence to equilibrium.